

Project 1

Secure Student Management System with Digital Audit Trail

Description

Develop a Flask/Django-based Student Management System with secure authentication, role-based access control, audit logging, and forensic evidence generation.

Features

- User Registration & Login
- JWT Authentication
- RBAC (Admin, Faculty, Student)
- Student Record Management
- Secure File Upload
- Activity Logging
- Forensic Audit Trail
- Dashboard Analytics
- CSV/MariaDB Storage

Security Features

- Password Hashing
- Session Management
- Input Validation
- CSRF Protection
- SQL Injection Prevention

Forensic Features

- User Activity Logs
- Login History
- File Access Logs
- Evidence Export

Learning Outcomes

- Flask/Django Development
- Database Security
- Authentication Engineering
- Digital Evidence Collection

Project 2

Cyber Incident Response & Threat Monitoring Dashboard

Description

Develop a Security Operations Center (SOC) dashboard capable of monitoring logs, detecting suspicious activities, and generating incident response reports.

Features

- Real-Time Log Collection
- Incident Detection
- Alert Management
- Threat Classification
- Dashboard Visualization
- Risk Scoring
- Incident Timeline

Security Features

- Secure API
- JWT Authentication
- Role-Based Access

Forensic Features

- Log Correlation
- Timeline Reconstruction
- Evidence Preservation

Learning Outcomes

- SIEM Concepts
- Incident Response
- Threat Intelligence
- Data Visualization

Project 3

Digital Evidence Management and Chain-of-Custody System

Description

Develop a digital evidence repository for cybercrime investigations.

Features

- Evidence Registration
- Evidence Storage
- Hash Verification
- Chain of Custody Tracking
- Metadata Extraction
- Evidence Search Engine

Security Features

- Secure Authentication
- Evidence Access Control
- Encryption

Forensic Features

- SHA256 Hashing
- Metadata Analysis
- Audit Logs

Learning Outcomes

- Digital Forensics
- Evidence Handling
- Legal Compliance
- Integrity Verification

Project 4

Secure File Upload & Malware Scanning Portal

Description

Develop a secure document management system that scans uploaded files and identifies suspicious content.

Features

- Secure Upload Portal
- File Validation
- File Type Detection
- Virus Signature Matching
- User Dashboard
- File Quarantine

Security Features

- File Upload Hardening
- Input Validation
- Secure Storage

Forensic Features

- File Hashing
- Metadata Collection
- Upload Logs

Learning Outcomes

- Secure Coding
- Malware Detection
- File Forensics

Project 5

Vulnerability Assessment and Web Security Scanner

Description

Develop a Python-based web vulnerability scanner for educational purposes.

Features

- URL Scanner
- Port Scanner
- Header Analysis
- SSL/TLS Validation
- Directory Enumeration
- Vulnerability Report Generation

Security Features

- Attack Surface Enumeration
- Risk Assessment

Forensic Features

- Scan Logs
- Assessment Reports

Learning Outcomes

- Web Security
- Vulnerability Discovery
- Security Assessment

Project 6

Secure Online Examination System with Proctoring Logs

Description

Develop a secure examination portal with anti-cheating mechanisms and forensic logging.

Features

- Student Login
- Exam Management
- Question Bank
- Timer System
- Result Calculation
- Activity Monitoring

Security Features

- JWT Authentication
- Session Monitoring
- Secure Question Delivery

Forensic Features

- Exam Activity Logs
- Login History
- Browser Monitoring Records

Learning Outcomes

- Secure Application Development
- Authentication Systems
- Audit Logging

Project 7

Mobile Device Forensics Reporting System

Description

Develop a forensic analysis platform for processing mobile device artifacts.

Features

- Device Registration
- Evidence Acquisition Records
- Metadata Analysis
- Report Generation
- Dashboard

Security Features

- Role-Based Access
- Evidence Protection

Forensic Features

- Mobile Metadata Analysis
- Evidence Integrity Validation
- Chain of Custody

Learning Outcomes

- Mobile Forensics
- Digital Investigations
- Reporting Standards

Project 8

Threat Intelligence and IOC Management Platform

Description

Develop a platform to collect, store, analyze, and visualize Indicators of Compromise (IOCs).

Features

- IOC Database
- IP Reputation Analysis
- Malware Hash Database
- Threat Feed Dashboard
- Threat Scoring

Security Features

- Secure APIs
- Authentication

Forensic Features

- IOC Correlation
- Threat Timeline Analysis

Learning Outcomes

- Threat Intelligence
- Cyber Analytics
- Threat Hunting

Project 9

Secure Healthcare Record Management System

Description

Develop a healthcare application with embedded security controls and forensic logging.

Features

- Patient Records
- Doctor Portal
- Appointment Management
- Secure File Upload
- Audit Dashboard

Security Features

- MFA Authentication
- Role-Based Access
- Encryption

Forensic Features

- Record Access Logs
- Incident Reporting
- Audit Trail

Learning Outcomes

- Secure SDLC
- Healthcare Security
- Compliance Requirements

Project 10

Security-Aware E-Commerce Platform with Digital Forensics

Description

Develop an e-commerce platform integrating cybersecurity and forensic capabilities.

Features

- Product Management
- User Registration
- Shopping Cart
- Order Processing
- Payment Simulation
- Admin Dashboard

Security Features

- Secure Authentication
- Session Security
- Input Validation
- API Security

Forensic Features

- Transaction Logs
- User Activity Tracking
- Fraud Detection Dashboard
- Evidence Collection

Learning Outcomes

- Full Stack Development
- Secure Coding
- Web Security Engineering
- Digital Forensics Integration

Recommended Industry-Level Capstone Projects

The goal is to demonstrate the maximum number of concepts from the curriculum, the strongest projects are:

Rank	Project
1	Cyber Incident Response & Threat Monitoring Dashboard
2	Digital Evidence Management and Chain-of-Custody System
3	Threat Intelligence and IOC Management Platform
4	Security-Aware E-Commerce Platform
5	Vulnerability Assessment and Web Security Scanner

These projects cover nearly all major topics from the syllabus including Python, Flask, Django, Databases, Authentication, Secure Coding, Vulnerability Analysis, Attack Surface Enumeration, Incident Response, Digital Forensics, Metadata Analysis, Evidence Integrity Verification, Linux Security, and Threat Analytics.